

*We acknowledge the use of OpenAI's ChatGPT as a supportive tool in drafting and refining portions of the text.

ENVR 210 2026W1: Environmental Systems and Spatial Analysis

This course introduces students to environmental systems thinking and the use of spatial analysis tools for examining environmental change across urban and natural landscapes. Students will explore topics such as ecosystem interactions, land-use change, climate impacts, geospatial technologies, and environmental decision-making through lectures, case studies, and applied exercises.

Course Learning Objectives

- Explain core environmental systems concepts and their interactions.
- Interpret spatial and geospatial datasets for environmental analysis.
- Apply environmental reasoning to evaluate real-world sustainability issues.
- Analyze environmental change using remote sensing and mapping approaches.
- Communicate environmental findings through written and visual formats.

Course Materials

- Required Textbook: *Introduction to Environmental Systems*, 3rd Edition.
- Weekly lecture slides and readings available on Canvas.
- Access to ArcGIS Online and QGIS software.
- Scientific calculator recommended for lab activities.
- Selected journal articles and case studies posted throughout the term.

Lecture Schedule

This schedule is subject to change.

Module	2026	Lecture
Environmental Foundations	Tuesday, September 8th	Introduction to Environmental Systems
	Thursday, September 10th	Ecosystem Components and Interactions
	Tuesday, September 15th	Climate Systems and Feedback Loops
	Thursday, September 17th	Human Impacts on Landscapes
Spatial Analysis	Tuesday, September 22nd	Introduction to GIS
	Thursday, September 24th	Map Projections and Spatial Scale
	Tuesday, September 29th	Remote Sensing Fundamentals
	Thursday, October 1st	Spectral Data and Image Interpretation
	Tuesday, October 6th	Spatial Data Visualization
	Thursday, October 8th	Midterm Review
	Tuesday, October 13th	MIDTERM

Environmental Applications	Thursday, October 15th	Forest Disturbance Mapping
	Tuesday, October 20th	Urban Heat Island Analysis
	Thursday, October 22nd	Watershed Monitoring
	Tuesday, October 27th	Wildlife Habitat Analysis
	Thursday, October 29th	Environmental Risk Assessment
	Tuesday, November 3rd	Climate Adaptation Strategies
	Thursday, November 5th	Environmental Policy and Data Ethics
	Tuesday, November 10th	Future of Environmental Monitoring
	Thursday, November 12th	Final Review

Breakdown of Marks

Participation and Discussion Activities: 10%

Labs and Assignments: 35%

Midterm Exam: 20%

Final Project: 10%

Final Exam: 25%